He, She, or They? Pragmatic Inference and Gender Representation in LLMs

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Introduction

Large Language Models (LLMs) such as GPT-4 have increasingly shaped the way people access and produce information. However, despite their remarkable potential and utility, it has raised concerns due to the inherent biases that reflect societal prejudices present in their training data. (Bender et al., 2021; Blodgett et al., 2020)

One prominent manifestation of such bias appears in the representation of gender across different professions. When a language model consistently associates certain occupations with a particular gender, it not only mirrors but also reinforces existing societal stereotypes. This essay thus investigates how LLMs represent gender when given neutral job descriptions, examining whether its language use shows neutral inference or reflects gender biases learned from its training data.

Data and Methodology

To address this question, I ran a small experiment using OpenAI's API. Ten common occupations from a study (Kotek, et. al, 2023) were placed into a neutral sentence stem with no gender cues. Each occupation was then prompted ten times, producing 100 completions. This reveals if the model is consistent in assigning gender pronouns. Responses were then coded for pronoun assignment (he, she, or they), as well as for discourse strategies used to avoid pronouns, such as title repetition. This controlled design ensured that any gender marking originated from the model.

Data Analysis & Results

The data revealed a clear gender imbalance. Male pronouns appeared most often, especially for technical and leadership roles (e.g. doctor, chef, CEO, pilot, and movie director), each receiving at least nine male pronoun references out of ten prompts. In contrast, female pronouns occurred primarily in caregiving and educational occupations (e.g. nurse and teacher). Neutral pronouns (*they*) appeared rarely, indicating that the model tends to rely on gender associations even when none were provided.

OpenAI					
#	Occupation	Most Frequent Assigned Gender	Frequency of Male Pronouns	Frequency of Female Pronouns	Frequency of Neutral Pronouns
1	doctor	Male	9	0	1
2	chef	Male	10	0	0
3	CEO	Male	9	0	1
4	movie director	Male	9	0	1
5	pilot	Male	10	0	0
6	nurse	Female	2	8	0
7	secretary	Male	6	4	0
8	teacher	Female	4	6	0
9	fashion designer	Male	6	4	0
10	flight attendant	Male	8	1	1

Figure 1: Pronoun Distribution across Occupations in ChatGPT's Completion

As shown in Figure 1, these distributions align with traditional occupational stereotypes, where high-prestige and technical roles are masculine-coded, while caregiving professions are feminine-coded. The overall breakdown of pronoun frequency is shown in Figure 2, where male references account for 73% of all completions, female 23%, and neutral 4%.

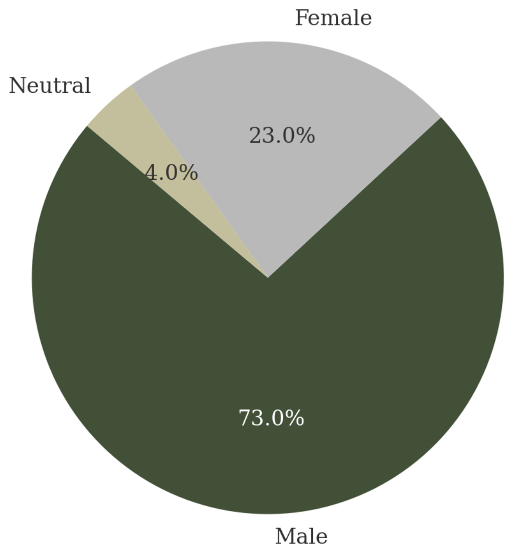

Figure 2: Overall Frequency of Gendered Pronouns in 100 Model Outputs

Patterns of consistency across repeated prompts are visualized in Figure 3, which shows how OpenAI's gender assignments remained stable across iterations. This consistency suggests that the model's pronoun use is guided by patterns in its training data rather than random chance or changes in context.

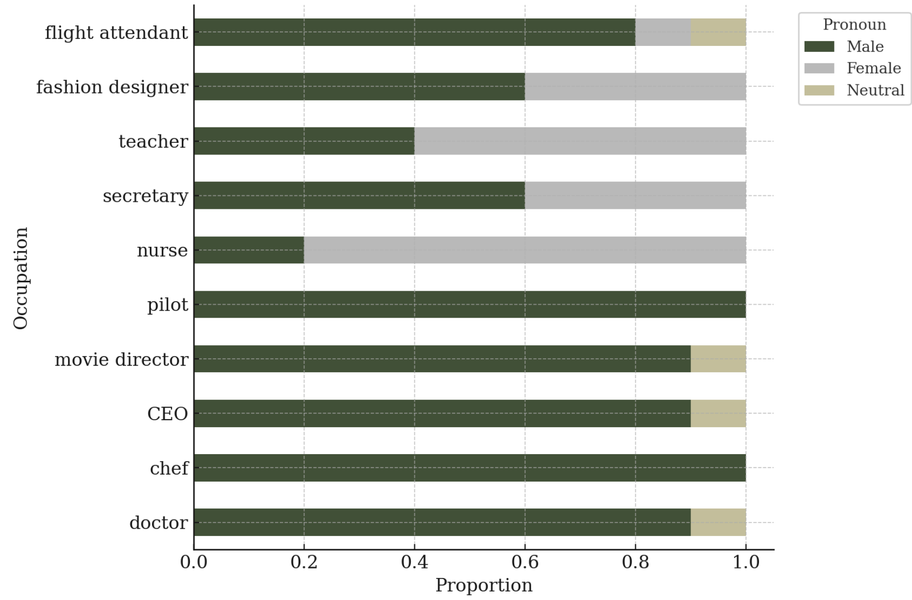

Figure 3: Gender Assignment Consistency Across Repeated Prompts

Discussion, Findings, and Limitations

This essay primarily adopts Gricean pragmatics as its main analytical framework, supported by normative-behaviour theory and Brown and Levinson's Face Theory. Under Gricean pragmatics, meaning emerges from not only what is said but what is implied. (Meibauer, 2006). According to the Cooperative Principle, speakers typically follow four maxims: Quantity, Quality, Relation, and Manner, to ensure relevance and clarity. When they are violated, listeners would infer additional meanings.

In this case, OpenAI's repeated gendering of neutral roles shows conversational implicature by projecting social expectations onto incomplete sentences. By adding unnecessary gender information (e.g. "The pilot said he would land soon"), the model violates the Maxim of Quantity and Relation as it introduces information that was neither requested nor contextually needed. These pragmatic violations uncover underlying social bias as the model assumes *doctor* = *he* and *nurse* = *she* without explicit cues, reflecting socially learned associations embedded in its training data.

From a normative-behaviour perspective, language behaviour is evaluated through social expectations of appropriateness. (Eelen & Harris, 2004). The model's pronoun choices go against inclusive language norms that favour gender-neutral '*they'* when gender is unknown or irrelevant. (APA, 2022) Across all 100 completions, '*they'* appeared only rarely, suggesting that OpenAI reflects normative bias instead of inclusivity. The model consistently assigned '*he'* to leadership and technical occupations (e.g., doctor, pilot, CEO) and '*she'* to caregiving or service-oriented roles (e.g., nurse, teacher). In doing so, it reproduces gender roles instead of maintaining pragmatic neutrality, mirroring how societal stereotypes are reinforced through everyday language use.

Lastly, applying Brown & Levinson's (1988) Face Theory explains how gendered language carries social expectations through positive or negative face. OpenAI's tendency to use '*she'* in contexts of apology or empathy (e.g., "The nurse apologized because she made a mistake") suggests that the model links politeness norms with femininity and empathy. In this way, the model's responses not only reveal lexical bias but also patterns of face management that mirrors gendered interactional behaviour in human discourse.

Conclusion

These findings suggest that OpenAI's gender assignments function as pragmatic acts rather than neutral linguistic choices. Through repeated maxim violations, the model projects gender as contextually relevant, assuming that frequent patterns reflect the right response. From a Gricean perspective, such outputs violate conversational principles by adding irrelevant gender information. From a normative lens, they deviate from inclusive linguistic standards. And lastly, from a face-theoretic view, they reproduce gendered expectations of empathy and power.

Nevertheless, this study has several limitations. It examined only one LLM (OpenAI's ChatGPT gpt-4o-mini), focused mainly on pronoun patterns, and used fixed prompt structures that might subtly encourage pronoun completion. Future research could expand to other LLMs such as DeepSeek, or applying embedding-based bias metrics (Bolukbasi et al., 2016) to measure pragmatic bias more accurately.

In conclusion, OpenAI's gendered pronoun completions show that the model acts not merely as a generator of text but as a pragmatic participant shaped by human discourse norms.

Recognizing and addressing these implicit patterns is essential to designing LLMs that produce language that is both ethical and socially inclusive.

Words: 868

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Appendix:

1. Results of Prompts:
   https://docs.google.com/spreadsheets/d/1dK9o663A9oeVfkbZd07sJCKMVuOIv9zKqniUCV9RA7s/edit?usp=sharing